

Code-switching in Mandarin and English Definite Noun Phrases

Introduction and Background: Code-switching in the nominal/DP domain has been extensively studied, particularly with regard to gender features of the determiner and the asymmetry between functional and lexical heads. Two main conclusions have emerged from this line of research. The Matrix Language Framework (MLF) (Myers-Scotton, 1993) states that the matrix language provides the determiner head, while the generativist approach (Cantone & MacSwan 2009; Liceras et al 2008) suggests it is determined whichever language specifies more phi-features. However, some features of the determiner, such as definiteness, have yet to be explored. Both definite noun phrases (e.g. *the boy*) and demonstrative noun phrases (e.g. *that boy*) can be used anaphorically, i.e. to refer back to previously mentioned discourse referents. Recent work has discovered different pragmatic constraints between anaphoric definites and demonstratives, finding crosslinguistic stability in them across many languages including English, Turkish, Bangla, and German (Saha et al. 2023), but noting Mandarin as an outlier in the semantics of its demonstrative (Saha et al. 2024). This study investigates the constraints on code-switching within anaphoric noun phrases by leveraging these differences between Mandarin and English demonstratives. We provide novel evidence supporting the MLF, which sheds light on the competition between different features of the determiner in code-switched utterances.

Methodology: We conducted three Mandarin-English and one English-Mandarin code-switching studies, each with contexts plus paired continuations following closely the design of Saha et al (2023)’s cross-linguistic monolingual studies. All studies consist of context and continuation sentences with the nominal code-switched into the embedded language. Participants (N=30 for each study) were asked to rate the two continuations provided in tandem (Figure 1), which varied according to whether the determiners (demonstrative/definite/bare) express the preferred features in Mandarin or English. Switched nominals were extracted from the SEAME corpus to motivate natural code-switching (i.e. 那个 *quiz* ‘that quiz’).

Results: When Mandarin is the matrix language, we observed patterns for code-switched English NPs similar to those found in the monolingual Mandarin context: in certain contexts Mandarin demonstratives are preferred over bare noun phrases (Figure 2), and bare nouns over English demonstratives (Figure 3) and English definites (Figure 4); the reverse is found for English matrix context code-switched into Mandarin, which mirrors the monolingual English context (Figure 5).

Discussion: The strong alignment between our results and those of Saha et al (2024) in finding that demonstrative noun phrases are preferred over bare NPs in anaphoric contexts when Mandarin is the matrix language and less preferred over English definites in English-Mandarin code-switching provides support for a primary role of the matrix language, not head noun, in determining the DP functional head. Moreover, we found that violating the definiteness feature of the matrix language is more acceptable than the violation at the language feature level. The comparison between bare and English demonstrative noun phrases further illustrates that having the matrix language dictate the language of the determiner is more acceptable than when only the definiteness feature is determined by the matrix language.

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作为上面给出的句子部分的延续，你可能认为第一个选项听起来比第二个选项更自然。如果是这样，对于第一个选项，您应该将滑块拖动到右侧某处；而对于第二个选项，您则需要将滑块拖动到左侧某处。

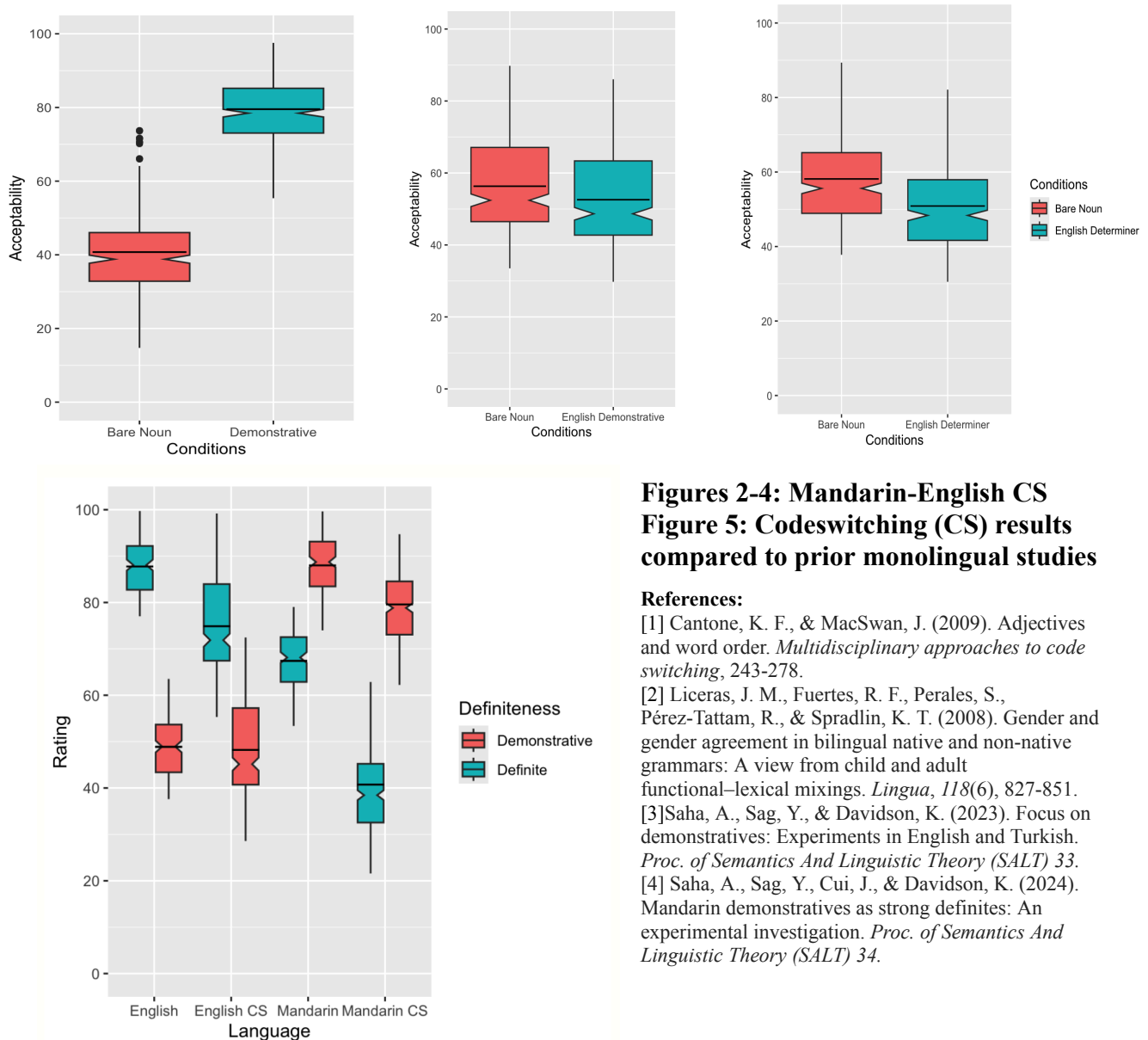
它们很难写。

● 最不自然 最自然

她们很难写。

● 最不自然 最自然

Figure 1: Sample Trial with Matrix Language Mandarin, noun in English



Figures 2-4: Mandarin-English CS

Figure 5: Codeswitching (CS) results compared to prior monolingual studies

References:

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